

A1 vs LRFShap — 7-dataset 결과 매트릭스

각 셀의 값 = $A1(r, sel\%) \text{ accuracy} - LRFShap(r, sel\%) \text{ accuracy}$ (단위 percentage point). 양수 = A1 우세, 음수 = LRFShap 우세. 같은 rank, 같은 selection % 에서 head-to-head, 둘 다 inv prediction 으로 평가.

MNLI (3-class) — 42/42 셀 A1 승

r \ sel	1%	2%	5%	10%	15%	20%	row avg
r=1%	+3.6	+9.0	+4.6	+4.2	+4.5	+3.3	+4.87
r=5%	+5.4	+6.8	+4.8	+4.9	+3.5	+1.2	+4.43
r=10%	+5.5	+8.0	+2.2	+2.6	+4.4	+1.9	+4.10
r=15%	+10.3	+4.2	+4.9	+2.7	+2.2	+0.7	+4.17
r=20%	+9.9	+5.6	+2.9	+2.5	+4.3	+1.8	+4.50
r=25%	+5.9	+6.3	+5.1	+3.4	+3.0	+1.0	+4.12
r=30%	+4.3	+4.9	+2.0	+5.3	+2.8	+1.9	+3.53
col avg	+6.4	+6.4	+3.8	+3.7	+3.5	+1.7	42/42 win

MRPC (binary imbalanced) — 35/42 셀 A1 승

r \ sel	1%	2%	5%	10%	15%	20%	row avg
r=1%	+0.5	+0.2	-3.4	-0.5	+0.7	+0.5	-0.33
r=5%	+1.7	+2.7	+2.5	+2.2	+3.2	+1.5	+2.29
r=10%	+3.9	+5.2	+4.4	+4.4	+3.9	+4.9	+4.45
r=15%	-0.2	+1.7	+5.1	+3.9	+6.1	+4.4	+3.51
r=20%	0.0	+1.5	+3.7	+4.9	+2.0	+3.7	+2.62
r=25%	0.0	0.0	+6.4	+5.4	+5.6	+5.1	+3.76
r=30%	-0.2	+0.5	+5.1	+5.4	+4.7	+5.9	+3.55
col avg	+0.8	+1.7	+3.4	+3.7	+3.7	+3.7	35/42 win

AG News (4-class) — 17/42 셀, low-budget niche

r \ sel	1%	2%	5%	10%	15%	20%	row avg
r=1%	+19.4	+6.0	-1.5	+0.7	+0.7	+0.9	+4.37
r=5%	+4.1	-1.5	+0.6	-0.6	-0.1	0.0	+0.42
r=10%	+9.2	0.0	-1.8	-1.8	-0.5	-0.1	+0.83
r=15%	+10.1	+2.1	-2.4	-0.8	-0.1	-0.3	+1.43
r=20%	+8.4	+0.5	0.0	+0.3	-0.3	0.0	+1.48
r=25%	+3.1	+1.2	-1.4	-0.8	-1.1	0.0	+0.17
r=30%	+3.9	+4.8	-0.8	-1.4	0.0	-1.3	+0.87
col avg	+8.3	+1.9	-1.0	-0.6	-0.2	-0.1	17/42

→ sel=1-2% column 모두 양수, 그 외 약간 마이너스. low budget niche.

SST-2 (binary hard) — 8/42 셀, very specific niche

r \ sel	1%	2%	5%	10%	15%	20%	row avg
r=1%	-6.3	-3.7	-0.1	-3.1	-3.8	-2.4	-3.23
r=5%	+0.1	-2.8	-3.3	-2.9	-3.4	-5.0	-2.88
r=10%	-1.9	-6.3	-4.5	-1.8	-2.2	-3.4	-3.36
r=15%	+6.5	+1.8	-1.4	-2.3	-2.6	-2.2	-0.02
r=20%	+2.9	+6.4	-0.6	-0.8	-1.3	-1.4	+0.88
r=25%	+7.8	+2.3	-2.8	-1.9	-1.4	-2.8	+0.21
r=30%	+12.5	-0.3	-0.6	-1.8	-1.0	-2.2	+1.09
col avg	+3.1	-0.4	-1.9	-2.1	-2.2	-2.8	8/42

→ high rank (r=15-30) × low sel (1-2%) 영역만 A1 우위.

MR (binary easy) — 4/42 셀

r \ sel	1%	2%	5%	10%	15%	20%	row avg
r=1%	-8.4	-4.4	-1.3	-1.5	0.0	+0.4	-2.53
r=5%	-0.3	-5.4	-3.2	-2.6	-2.0	-1.0	-2.42
r=10%	-8.5	-13.8	-3.1	-2.7	-2.3	-3.3	-5.62
r=15%	+1.8	-3.3	0.0	-2.1	-0.9	-2.6	-1.18
r=20%	-1.2	-3.3	-2.3	-0.7	-0.9	-2.0	-1.73
r=25%	-2.9	-3.9	-2.9	-1.8	-1.3	-1.4	-2.37
r=30%	+0.1	+1.5	-0.1	-0.4	-2.4	-2.7	-0.67
col avg	-2.8	-4.7	-1.8	-1.7	-1.4	-1.8	4/42

QQP (binary marginal) — 7/42 셀

r \ sel	1%	2%	5%	10%	15%	20%	row avg
r=1%	+0.7	+2.9	+0.6	+2.5	+1.5	+1.3	+1.58
r=5%	-0.3	-0.8	-3.4	-1.6	-0.9	-0.5	-1.25
r=10%	-3.2	-4.4	-2.9	-2.3	-1.6	-2.6	-2.83
r=15%	-4.5	-6.6	-5.6	-2.5	-2.8	-3.5	-4.25
r=20%	-0.4	-8.4	-3.0	-2.9	-2.6	-2.9	-3.37
r=25%	+1.6	-3.1	-3.4	-3.7	-3.8	-2.2	-2.43
r=30%	-0.4	-8.8	-5.4	-1.7	-2.0	-1.6	-3.32
col avg	-0.9	-4.2	-3.3	-1.7	-1.7	-1.7	7/42

→ r=1% row 만 살짝 우위 (+1.6 avg). 나머지 패배.

RTE (binary, kernel mismatch) — 4/42 셀

r \ sel	1%	2%	5%	10%	15%	20%	row avg
r=1%	-14.1	-7.6	+2.5	+0.7	-2.5	-1.8	-3.79
r=5%	-5.1	-7.2	-9.0	-4.3	-6.5	-5.8	-6.32
r=10%	-2.2	-4.3	-8.7	-5.8	-9.4	-9.8	-6.68
r=15%	+2.9	-6.1	-2.2	-5.8	-8.3	-9.4	-4.82
r=20%	-0.7	-5.4	-5.0	-5.4	-8.3	-8.3	-5.54
r=25%	+4.0	-2.2	-7.2	-7.2	-9.4	-8.3	-5.06
r=30%	-2.9	-2.2	-4.7	-7.6	-10.5	-9.0	-6.14
col avg	-2.6	-5.0	-4.9	-5.1	-7.8	-7.5	4/42

→ 모든 영역 일관 패배. 격차 sel 가 클수록 확대.

7-dataset 요약

Dataset	Class	wins/42	row avg 범위
MNLI	3-class	42/42	+3.5 ~ +4.9
MRPC	binary imbalanced	35/42	-0.3 ~ +4.5
AG News	4-class	17/42	+0.2 ~ +4.4
SST-2	binary hard	8/42	-3.4 ~ +1.1
QQP	binary	7/42	-4.3 ~ +1.6
MR	binary easy	4/42	-5.6 ~ -0.7
RTE	binary mismatch	4/42	-6.7 ~ -3.8

Setup: n=2000 (RTE 만 2490), seed 2026, TMC 500 iter, $\rho=1e-2$ (eigen), $\lambda_{inv}=1e-6$.